

Crowd Counting and Density Estimation using Deep Network: A Comprehensive Survey

Muhammad Jawad Babar, Malik Muhammad Saad Missen, Mujtaba Husnain, Ali Samad, Muhammad Nasir and Abdul Karim Nawaz Khan

Abstract—In machine vision, the tasks of both the crowd counting and the crowd density estimation attracted a number of researchers from different research institutes throughout the world. These tasks are of significance importance while analyzing the surveillance videos and images both at the real and offline modes. A number of different conventional and modern algorithms are being applied to estimate the accurate amount of people from the crowd from the target image or video file but failed to provide the required results due to heterogeneity of the data instances. To resolve the issue, machine learning models like deep networks like Convolution Neural Network (CNN) and its variants were introduced. In this paper, a detailed and comprehensive review of the work related to both of the crowd counting and density estimation using deep network is provided in quite a very unique way to make it useful for the researchers working in the field of computer vision. Following are the salient features of our review that play key role in making our contribution worthwhile and unique among the surveys and reviews of similar kind: (i) our survey intends to classify the noteworthy literature with respect to the application and the tasks related to crowd counting and density estimation, (ii) the survey also tried to cover in-depth contributions in crowd counting existing works at different level like images or videos (iii) this state-of-the-art review covers each article in the following dimensions: the designated task performed, source level, results obtained, features used, and (iv) lastly it concludes the summary of the related articles according to the publishing years, related tasks (or subtasks), and types of classifiers used. In the end, major challenges and tasks related to crowd counting and density estimation in computer vision are also discussed.

Index Terms—Crowd counting, Convolution Neural Network (CNN), Deep Network.

I. INTRODUCTION

IN the field of computer vision, the crowd counting refers to the task of calculating and estimating the number of persons in image (or video file), in a given area or space at a particular point in time [1]. On the other hand, the density estimation deals with crowd counting to estimate the number of people in an image or video frame. It does so by creating a density map that assigns a density value to each pixel in the image or a certain area of the image [1], [2], [3]. The density map is then used to estimate the total number of people in the image by counting the number of pixels with high density values. The algorithms and models based on computer vision techniques play key role in the in depth analysis of the image (or video) file captured by cameras [4], [5], [6].

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The primary purpose of crowd counting is to provide a near to accurate estimates of the number of people in a crowd or gathering. This activity assist us in different contexts like event planning, crowd management, and security. Crowd counting algorithms typically use machine learning models to detect and track individuals in the images or video, and then use statistical techniques to estimate the total number of people present. Crowd counting can be a challenging task, particularly in situations where the crowd is dense, or the lighting or camera angles are suboptimal. However, advances in computer vision and machine learning have led to increasingly accurate and efficient crowd counting algorithms in recent years[7], [8], [9], [10].

A number of authors concluded that the research framework for crowd counting typically involves the following steps [11], [1], [12], [13]:

- Dataset collection: Collecting a dataset of images or videos with ground truth annotations that indicate the number of people present in each frame.
- Preprocessing: Preprocessing the images or videos to remove noise, resize the images, and normalize the pixel values.
- Feature extraction: Extracting features from the images or videos that can be used to identify and count people. This can involve using techniques such as convolutional neural networks (CNNs), local binary patterns (LBPs), or scale-invariant feature transform (SIFT).
- Counting algorithm: Using a counting algorithm to estimate the number of people present in each frame based on the extracted features. This can involve using techniques such as regression-based methods, density-based methods, or detection-based methods.
- Evaluation: Evaluating the performance of the counting algorithm on a separate validation set or using cross-validation. This can involve using metrics such as mean absolute error (MAE), mean squared error (MSE), or root mean squared error (RMSE).
- Improvement: Improving the performance of the counting algorithm by refining the feature extraction, counting algorithm, or both.
- Application: Deploying the crowd counting algorithm in real-world applications such as video surveillance, crowd management, and event planning.

Overall, the research framework for crowd counting involves collecting and preprocessing data, extracting features, designing and evaluating counting algorithms, and deploying the algorithm in real-world applications.

In this paper, we present the state-of-the-art survey on recent advancements in crowd counting and density estimation that focused on improving the accuracy and efficiency of existing methods, as well as developing new techniques to address specific challenges using deep learning approaches like the Convolution Neural networks (CNNs) and its variants. More advanced architectures, such as the fully Convolution Network (FCN) [14] have also been covered to better capture the spatial relationships between different regions of the image and improve accuracy.

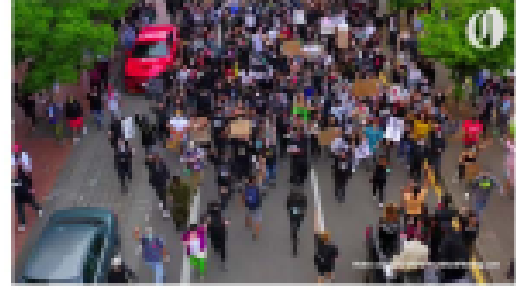
Another important trend in crowd counting is the use of multi-view and multi-modal data [15], [16], [17], [18], [19]. The researchers have explored the use of multiple cameras and sensors to capture different perspectives and modalities of a crowd, such as depth, thermal, and audio data [20], [21], [22], [23]. By fusing these different sources of information, it is possible to improve the accuracy and robustness of crowd counting systems. To evaluate the models for crowd counting, the Mean Absolute Error (MAE) and Mean Squared Error (MSE) are used that better capture the performance of on real-world datasets.

Crowd counting has various applications in different fields, including:

- **Public safety:** Crowd counting can help law enforcement agencies estimate the number of people in a particular area during a public event, such as a protest, rally, or concert [24], [13]. This information can be used to allocate resources and ensure public safety.
- **Shopping Malls Management:** Shopping malls and retail stores can use crowd counting to analyze foot traffic and customer behavior [25], [26]. This can help store managers optimize store layout, staffing, and marketing strategies.
- **Transportation:** Crowd counting can be used in transportation planning to estimate the number of people using public transportation at a given time [27], [28]. This information can be used to optimize public transportation routes and schedules.
- **Tourism:** Crowd counting can help tourism authorities estimate the number of visitors to a particular attraction or event [29], [30]. This can help them plan and allocate resources accordingly.
- **Town (or city) planning:** Crowd counting can be used to estimate the number of people in a particular area at different times of the day [13]. This information can be used to plan and design public spaces, such as parks, plazas, and streets.
- **Disaster management:** Crowd counting can be used in disaster management to estimate the number of people affected by a disaster, such as a hurricane or earthquake [31], [32]. This information can help emergency respondents in allocating the resources and prioritize rescue efforts.

Figure 1 depicts the crowd counting process by both the detection and density estimation process.

Moreover, there are also a number of tasks that are relevant to crowd counting namely object counting such as vehicles in transportation [33], [34], [12]; cell counting in medical



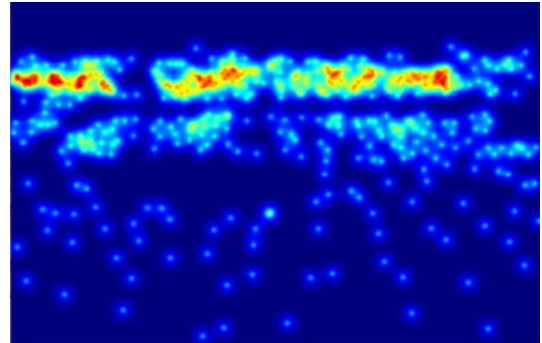
(a) [



(b)



(c)



(d)

Fig. 1: Examples of crowd counting methods: (a) the image source for detecting the crowd, (b) result of the crowd counting using detection method, (c) the source image for crowd counting using density estimation method, (d) the result obtained by density estimation

image analysis [35], [36] scene understanding in video analysis etc [37], [38], [39].

In this survey, we performed an in-depth analysis on the

noteworthy work related to the crowd counting and density estimation methods used on the CNN and its variants. Moreover, we also discussed conventional approaches that were made into use for the said purpose. In the previous literature review of same kind, the survey articles [40], [41], [42] presented the analysis of the conventional approaches only. Similarly, some other related survey articles reported in [43] discussed only the generic role of the approaches in crowd counting and also covered the papers published in last 6 to 8 years only. In [7], the authors tried to cover the articles that presented the task of crowd counting using single image only. Besides all these, we aimed to present a review in a very systematic and disciplined way of the noteworthy articles on crowd counting and density estimation using CNN and its variants. Furthermore, the details about the datasets, the details about the journals and the striking conferences is also given.

Our paper is organized as follows: Section II discuss the conventional methods of crowd counting and density estimation. The Section III covers the recent CNN-based models for crowd counting in a very comprehensive and disciplined way while the details about the benchmark datasets used in the work of crowd counting and also the key journals that are being targeted by the researchers for publication is given in Section IV. In Section V, we summarized the work with future trends and issues to be addressed.

II. CONVENTIONAL METHODS FOR CROWD COUNTING

In this section, we discussed about the conventional and traditional approaches used in crowd counting. Since we aim to present the survey on the CNN-based approaches for crowd counting, this section will briefly present the conventional approaches.

A. Regression based methods

In [44], the authors presented a novel approach to counting the people in an unsupervised target set by projecting the bi-knowledge from regression- and detection-based models in a supervised source collection. The proposed method leverages the complementary strengths of these two models to capture different modalities of the crowd distribution. Similarly, in [41], the authors explored different strategies for population profiling and density estimation in public spaces using video file. The authors have evaluated various techniques, including counting by detection, clustering, and regression, and have highlighted the importance of individual components in the processing pipeline. An array-based infra-red detection method for crowd existence is presented in [45]. The detection and tracking algorithms are capable enough to give significant performance that might be expected from a 16×16 array when allied with interpolation techniques. Therefore, it seems that array-based infra-red detection technology is a promising alternative to traditional methods of people counting and tracking. In some noteworthy work reported in [46], [47], [47], [48], [49], [50], [44] used different variants of regression methods to detect and count the people in crowd using single image. Similarly, the histogram based methods are observed in the noteworthy work reported in [51], [40], [52].

In [53], Markov Random Field- (MRF) based method is used for crowd monitoring by analyzing the metro station video clips having crowded passengers clusters moving in different directions. A video-based analysis for crowd counting is performed in [54] in which a mixture of algorithms are applied for different tasks like low-level background updating; detection of change in object movement; tracking and labeling the active and static areas in the image; and finally retrieving the features of each detected area; Then image-understanding algorithms are made into use to estimate the number of people on the basis of the information obtained through the algorithms in former step.

Similarly, experimental work using a number of regression-based models are applied for performing tasks related to crowd counting and monitoring like crowd estimation in labeled images data [55]; adapting the models using Bayesian regression methods [56] and associate the patch features and the regions of the localized target moving objects [3]; counting the persons using Residual Regression [57]; crowd density estimation using sparse linear regression [58]; crowd counting using weekly-supervised based approach of Multiple Auxiliary Tasks Training (MATT) [59]; counting using Hierarchical adaptive regression [60].

B. Detection based methods

In [61], the proposed method is specifically designed for detecting human heads in crowded environments using depth information and 3D scale-adaptive filtering. The limitation of the proposed method lies in the fact that the authors did not mention its applicability to detect other body parts besides the human heads. The authors in [16], designed a method for human detection in crowded scenes from static images. The model detects an individual human as an assembled structure of natural body parts and introduces edgelet features, that are considered as an advanced type of silhouette-based features. The model then learns the other features using a boosting method and combine their responses to form a joint likelihood model that consists of the data of more than one inter-occluded human impressions. The human detection problem is formulated as maximum a posteriori (MAP) estimation, and show promising results on the state-of-the-art datasets. The authors claim the it is possible to extend the proposed method to real-time video streams also.

A quite different approach is used in the work reported in [62] that is based on Histogram of Oriented Gradients (HOG)-based descriptors. The HOG descriptors works better in robust visual object recognition, specifically in human detection. A series of experiments conclude that HOG-based descriptors provided significant results by extracting the key features for the same task. Figure 2 depicts the proposed feature extraction chain used in the work.

The authors, in [63], make use of unsupervised data-driven Bayesian clustering method for detecting the individual entities in crowds. The method is capable enough to track the simple image features and groups them into clusters representing independently moving entities, without the need for subject-specific models or supervised learning.



Fig. 2: An overview of feature extraction and object detection chain. The detector window is tiled with a grid of overlapping blocks in which Histogram of Oriented Gradient feature vectors are extracted. The combined vectors are fed to a linear SVM for object/non-object classification. The detection window is scanned across the image at all positions and scales, and conventional non-maximum suppression is run on the output pyramid to detect object instances, but this paper concentrates on the feature extraction process [62].

The unsupervised Bayesian clustering algorithm differs from previous approaches to detecting individuals in crowds by focusing on motion as the primary source of information. While other methods often rely on subject-specific aspects. A statistical approach is reported in [64] aim to detect the people passing through a gate of metro station, thus achieving accuracy of 93%. Quite similar work is also performed in [65], [17] in which authors present an algorithm that focuses on local regions of an image to learn mid-level features that discriminate between pedestrian and non-pedestrian classes. By using AdaBoost [66], [67], [16] to select a subset of learned shapelets, the algorithm is able to retrieve more useful information and achieve significant results. The segmentation approach is applied in the work of [18], [68], in which the number of people are estimated. The authors have combined two algorithms to provide an accurate estimation of people counts in surveillance scenes with people gathering and waiting. The experiments conducted demonstrate the effectiveness of their approach.

In [69], [70], [71], [72], an auto-segmentation-based approach is used to extract the human silhouettes using inverse transform as well as median filter reducing the cost of computing and handling various complex monitoring situations. The experiments performed on both the self-made datasets and state-of-the-art datasets like PETS2009, ShanghaiTech [73], TRaffic AND COngestionS (TRANCOS) [74], and UCF-QNRF [75]. Authors in [76], [77], [78], [79], [80] make use of different detection methods to perform crowd counting related tasks on different datasets.

A hybrid approach is applied in [81] for estimating number of people at image level. The Haar wavelet transform (HWT) [82] is used to extract the features of the contour of a head structure using the promising features depicting the contour of a head. Then the wavelet template assisted in categorizing the different heads in the crowd. Finally, Support Vector Machine (SVM) [83] is used as a classifier to classify the extracted features of the head-like contour as either a head or not. The clusters of the people having same pattern of movement and displacement are categorized during offline processing of the images in [15]. The number of people in these clusters is estimated by their respective sizes by carefully reading the pattern of merging and splitting.

The proposed system in [84] includes two different modules: image processing and image understanding. Data obtained through the image processing is a kind of data curves corresponds to each different image thus depicting the the region

of interest (ROI). In second module, our system is based on the Radial Basis Function (RBF) [85] network is used as classifier. The key features of this system like the effectiveness, flexibility, and ability to perform accurate people counting in real-time using unconstrained visual-based images make this work noteworthy among the published work os the same kind in early 2000. Similarly, the vision-based people counting system described in [86] in which the authors used a fixed single camera to detect and track moving people. The camera is hung from the ceiling of the gate to prevent the image data of the passing people along various directions, to be not fully overlapped. The proposed system is made capable enough to detect the people passing by even they are partially overlapped, by drawing the bounding box around each detected person. The approximated convex hull of each individual in the tracking area is obtained to provide more accurate tracking information by indicating a threshold value if the bounding boxes are overlapped.

The system in [87] tried to extract foreground objects from the background using each sensor and sends the resulting bitmaps (i.e., silhouettes) over the network. The data from all sensors is then aggregated to compute a planar projection of the visual hull of the whole scene. This projection is used to bound the number and possible locations of people, a non-trivial task given that portions of the visual hull may in fact be empty. The visual hull of a scene is the intersection of the projected silhouette cones from all cameras. The planar projection of the visual hull is used to bound the number and possible locations of people in crowded environments. The system tracks regions of space that are determined to be occupied, and a geometric algorithm calculates bounds on the number of persons in each region of the projection, after phantom regions have been eliminated. Our approach is not based on the explicit detection of people in the images, so there is no pairwise matching of people across frames. Furthermore, the authors claim that the proposed system is capable to work in real-time even on any arbitrary wireless sensor network. A quite similar work for monitoring the crowd is observed in [88] in which motion-based estimation map system is proposed in which the region of interests (ROIs) are then segmented and analyzed for noise removal. The crowd and other related objects are then detected using optical flow and background removal on every frame. All the detected entries are tracked in the next phase followed by classifying the different objects on the information gathered about the different objects like po-

sition, velocity, acceleration/deceleration, bounding rectangle, and shape features.

The algorithm in [89] uses a target detecting algorithm based on projection of gray-scale image to detect targets. According to authors, the proposed method is able to minimize the effect of occlusion that is also regarded as one of the challenging problems to date in accurate detection of overlapped objects; using template matching. Furthermore, the proposed model is found successful in retrieving the key features related to the shape and color distribution. The authors in [90], aimed to explore the association among the geometric interpretation of the silhouette images of the objects and the background thus proving the fact that this association can be applied formally to all the foreground pixels. The model is made efficient for application of object detection in real time also.

The key contribution of the authors in [91], is the normalization of viewpoint invariant features retrieved for crowd counting by analyzing single image. These features include edge orientations and blob size histograms used in the training data. The authors also claim that these features play key role in detection of the objects than the simple features in terms of handling noise and accurately counting pedestrians moving in clusters. A novel approach to partition the densely crowded semi-rigid objects by clustering tracked feature points is observed in [92]. Kanade-Lucas-Tomasi (KLT) tracker [93] is used with some modifications for feature extraction that is used to process video into a set of feature trajectories. This set of trajectories provides a substrate for motion analysis. The KLT tracker helps with motion analysis of crowded objects by allowing for the tracking of individual features even in the presence of occlusions and other challenges posed by crowded scenes.

The main contribution of [94] in people detection walking on pedestrian is to detect and classify the points lying on a connected Riemannian manifold [95] using the geometry of the space, that ultimate results in generating the superior detection rates compared to previous approaches. The model used in [72], [96] is hybrid of two stochastic models namely Markov Chain Monte Carlo framework and Bayesian processes is used to efficiently search for the maximum a posteriori configuration of shapes to detect the object within a bounding box. The experimental work proved the significance of the hybrid approach in counting the moving persons while analyzing a surveillance video both in offline and online mode.

The authors in [97], [70] proposed novel approach for detection and segmentation of walking people using hierarchical part-template matching approach by combining local part-based and global shape-template-based schemes. For evaluation purpose, five datasets namely, The National Institute for Research in Digital Science and Technology (INRIA), MIT-CBCL, and USC-B including Caviar Benchmark and Munich Airport data sets, were tested achieving significant results. Figure 3 shows the example of detection process with and without background.

A probabilistic approach to localize multiple people using multiple calibrated camera views is described in [19], [98], [99] in which the a number of images were masked using motion masks at different orientations. This masking process

assist in localizing the pixel-level feature accurately specially the moving body parts i.e. legs and arms in to and fro motion. Random forest based method is used in the work of [100] in which the low-level feature to the people count are retrieved. In order to preserve the characteristic features, dimensionality reduction approach is used to reduce computation time and space.

Similarly, the crowd counting using other detection methods like feature mining for region based crowd counting [101], [102], [103]; feature points detection to predict the flow of crowd flow [104]; analyzing the cluster by tracking the occlusions with spatial and temporal constraints using hierarchical tree structures, finding coherent regions and clusters of persons among different clusters [105]; crowd estimation using label distribution learning [106]; crowd monitoring using agent-based modeling [107]; estimating the persons in crowd using conditional marked point process [108]; crowd monitoring using via Density estimation based visual object counting (DEVOC) [109]; using partial annotation of image for crowding.

III. CNN-BASED CROWD COUNTING METHODS

The Convolutional Neural Networks (CNNs) are commonly used for crowd counting because they can effectively extract features from images, and are able to learn and represent complex patterns in the data. Crowd counting is a challenging task because of the large number of objects that need to be counted, and the high variability in their appearance due to occlusions, scale changes, and perspective distortion. The key reason in using CNN in crowd counting and crowd density estimation tasks lies in the fact that CNN structure is able to learn and the nonlinear relations within the data. It is more accurate to use CNN to learn the nonlinear relationship between images and the number of people in images or their corresponding density map. Figure 4 depicts the method of our survey.

A. Crowd counting at image level

In this section, the noteworthy work published related to the task of crowd counting performed by analyzing the crowded scene images, is covered in detail.

According to the noteworthy work reported in [5], [6], [110], the researchers proposed deep CNN regression model outperforms existing methods that rely on hand-crafted features or auxiliary detectors. Extensive experimental results performed on UCSD [46], UCF-CC [50], PASCAL VOC 2012 [111], PETS-2009 [50] show that the methods used succeed in achieving the significant performance in terms of the mean and variance of absolute difference compared to state-of-the-art methods.

A specifically designed model namely, CrowdNet, based on CNN is proposed in [112], is a game-changer for crowd counting. With its combination of deep and shallow, fully convolutional networks, it effectively captures both high-level semantic information and low-level features necessary for crowd counting under large scale variations. The proposed model was tested on the challenging UCF-CC 50 dataset, and

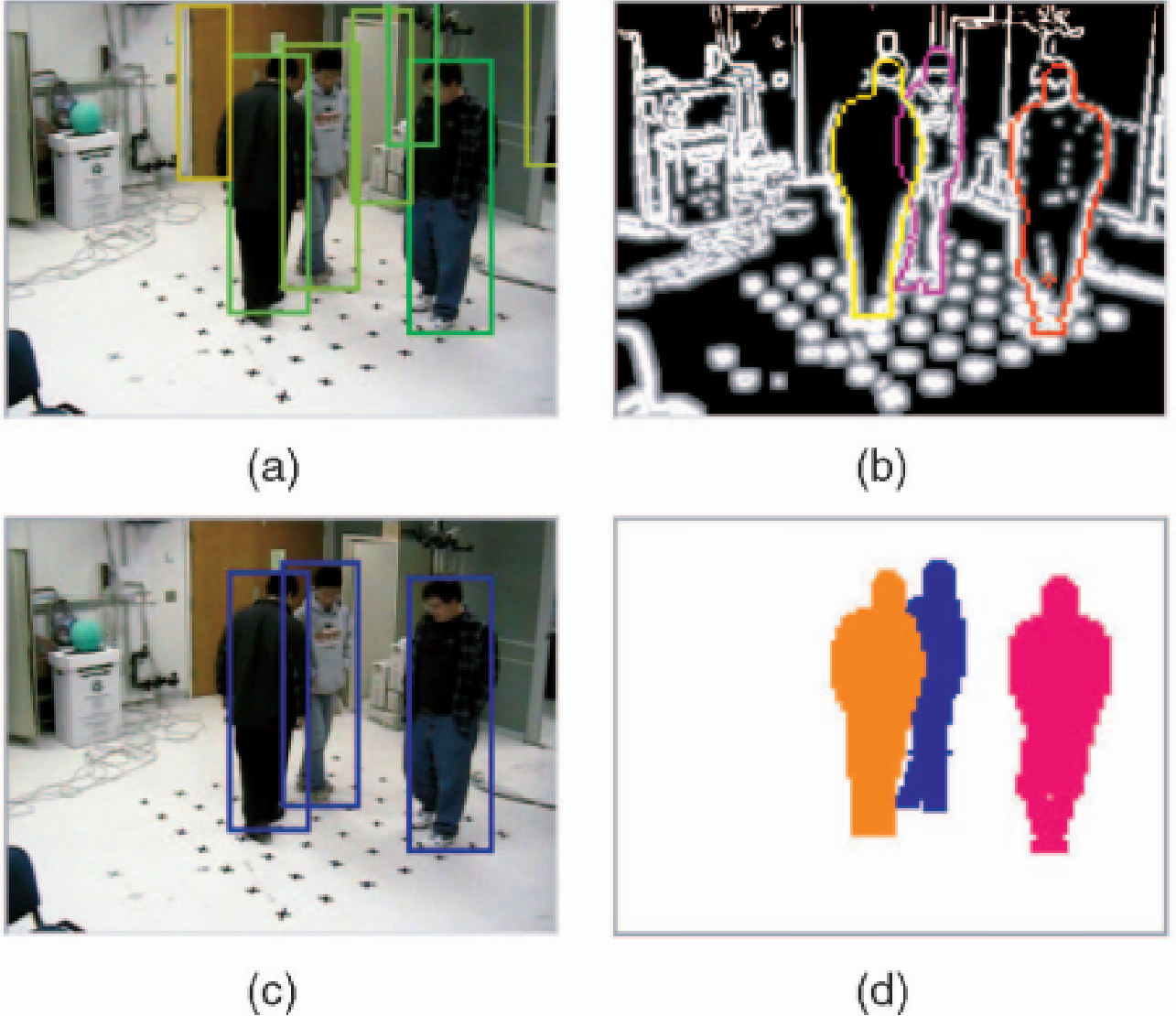


Fig. 3: Example of detection process without background subtraction. (a) Initial set of human detection hypotheses, (b) human shape segmentation, (c) detection result, and (d) segmentation result (final occlusion map [97]).

TABLE I. Quantitative results of the approach used in citeboominathan2016crowdnet along with other state-of-the-art methods on UCF-CC 50.

Method Mean	Absolute Error (MAE)
Learning to Count [113]	493.7
Density-aware Detection [114]	655.7
FHSc	468.0
Density-aware Detection [115]	467.0
Proposed [112]	425.5

shown to outperform the state of the art methods. Table I refers to the results obtained by performing the experiments on the dataset aforementioned.

Similarly, in [116], the authors proposed a novel approach to counting the number of people in surveillance videos using a deep Convolutional Neural Network (CNN). The novelty lies in the fact that unlike others [117], [118], [119] only make use of

the numbers of crossing-line pedestrians as supervision, while in this work the authors considered that in how much time the Line of Interest (LOI) is crossed by a specific pedestrian. This rich supervision information helps the CNN in learning more discriminating and key feature representations.

The locality identification of Region of Interest (ROI) in a video or an image play key role in crowd monitoring related activities including regression problems. This concept is used in a quite different work reported in [120], [35], in which variant of Recurrent Neural Network (RNN) namely, Low Short Term Memory (LSTM) network is used for generating the sequences and dependencies to train the model. It is also claimed that the proposed model differs from current people detectors in that it is based on decoding an image into a set of people detection. Instead of scanning an image in a sliding window fashion or classifying a discrete set of proposals, the proposed model generates predictions jointly and directly

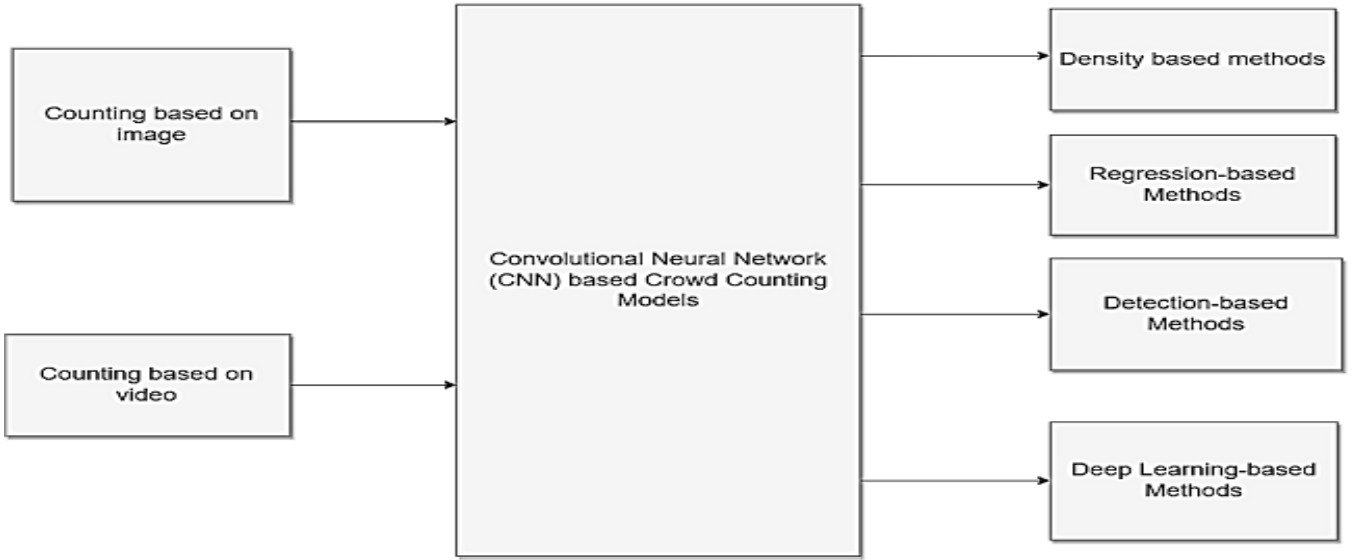


Fig. 4: Categorization of crowd counting method based on CNN.

outputs a set of distinct detection hypotheses. This approach is quite useful in making the common post-processing steps like the non-maximum suppression, redundant and unnecessary. on sets of detection. An effective Multi-column CNN (MCNN) model is designed by [73], [121], [122], [123], [124] to generate the image contents to its corresponding density map regardless of the size and resolution of the image. The MCNN is designed in a way that is capable of detecting the variations in the head/person size using variable perceptive filters.

A simple but quite effective online hard example mining (OHEM) algorithm is proposed for training a region-based CNN (R-CNN). As a result, these examples make the training process more effective for detecting the objects and persons by analyzing the single image. An automatic selection of these hard examples can make training more effective and efficient. The proposed model of OHEM was evaluated and achieved results of 78.9% and 76.3% maximum APosteriori (mAP) on PASCAL VOC 2007 and 2012 respectively. Similar model is applied for density estimation in the work of [125], [126] in which a novel end-to-end interleaved network of CNNs is modeled to learn both the crowd counting and density map estimation. A series of experiments performed on standard datasets show that the model succeeds in achieving the lower error rate while counting number of person and also acquiring quality density maps.

The Contextual Pyramid CNN (CP-CNN) for generating the quality-based the density estimation of the images having crowded people. Furthermore, the global and local contextual values are made into use for computing the count of the crowd. According to the authors, the proposed CP-CNN is comprised of four modules: Global Context Estimator (GCE), Density Map Estimator (DME), Local Context Estimator (LCE) and a Fusion-CNN (F-CNN). It is worthy to note that CGE is based on variant of CNN namely, VGG-16, specifically designed for image and video analysis. This component encodes the global

contextual features and is trained to map the images to their corresponding density classes. A deeply-recursive network (DRResNet) based on deeply convoluted blocks is used by a group of researchers [24] for crowd counting. The inherent structure of the network is deeper and dense thus keeping the number of parameters unchanged that improves the ability to capture statistical features from the image in the context of the crowd. Similar work can be seen in [127] with the difference that it uses a novel training loss function by integrating the Euclidean loss with the structural similarity loss. attenuate the blurry effects of density map estimation

The authors in [128] make use of an Adaptive Counting Convolutional Neural Network (A-CCNN) to improve accuracy for counting the people in crowd counting. The authors achieved significant results on the three widely used challenging datasets, namely the UCSD, UCF-CC and the crowd dataset collected by the authors from a metro station [129].

Loss functions help in evaluating the performance of the algorithm used for evaluating a specific dataset. It is computed from the tune-able parameters of the specific algorithm used. A comprehensive work on composition of loss functions is observed in [75], [130] when estimating the density map for the dense crowd. They proposed a novel approach for resolving the issues related to crowd counting, density map estimation and exploring the localization and neighborhood of the people in a given dense crowd image. These three problems are related to each other that the loss function for one of these tasks affects the optimization of the solution. To resolve this critical issue, the authors proposed some useful evaluation metric, Composition Loss, that can be used for training neural network. Then the loss function is compared with the results obtained from the other deep CNN networks, specifically designed for crowd counting. Furthermore, they also created a large-scale state-of-the-art dataset named UCF-QNRF.

The details about the datasets used for evaluation of crowd counting is given in Table II.

The counting of people in crowd image captured at cross-scene scenario is a complex task when background is changed in new images while training the model. This issue is addressed in [135] using a cross-scene counting model in which the model is learned with information transferred from other scenes. The features are explored from the manually segmented foregrounds in crowd images. The counting model were then developed using domain adaptation-extreme learning machine (DA-ELM) that is effectively trained via iterative optimization with training samples of both domains. In [136], the authors make use of image pyramid to resolve the issue of scale variation instead of using kernel filters of variable size and dimensions. Furthermore, the authors aimed to embed the predictions from different scales (using adaptively changing per-pixel weights) making the method more adaptable to scale changes within an image.

In machine vision, a novel model namely, Generative Adversarial Network (GAN) is categorized of applied machine learning models that play role efficiently in generative Artificial Intelligence (AI). Similar model is proposed for crowd counting in [137], [138], [139], [140] that performed better by lowering the error rate caused by different scales of crowd using full advantage of GAN deep model that holds a fully connected architecture. To resolve the common issues of scale and perspective changing in crowd images, the authors in [141] applied scale-adaptive CNN (SaCNN) model that is also considered as the mainstay of fixed receptive fields of small sizes. A series of experiments were performed on ShanghaiTech, UCF-CC-50 and WorldExpo'10 and SmartCity datasets and the results provide the significant results by introducing the two loss functions to jointly optimize the model.

In Machine Learning, the generalizable features refer to features or patterns in the data that are relevant for the task at hand, but are also likely to be useful for other related tasks or datasets [142]. These features are often identified through feature engineering or learned automatically through deep learning algorithms. of SaCNN over the state-of-the-art [143]. This phenomenon is well explained in [144] in which deep negative correlation learning (NCL) is applied to improve the ability to generalize the features to make the learning process robust and fast. On the basis of this model, they designed model, Decorrelated ConvNet (D-ConvNet) having the capability of spotless generalization by using the the inherent characteristics of the model. Following the model of CNN, the authors in [145] make use of Visual Geometry Group (VGG16) [146] model by eliminating the fully connected layers. Furthermore, the global and local features of the crowd image were fused using the pyramid network to represent the better constancy.

A data-driven and deep learning method Congested Scene Recognition (CSRNet) is observed in the work of [147], [148] in which the congested crowd image is analyzed to generate high quality density map. there are two components of proposed model. At front end, for 2D feature extraction, the conventional CNN is made into use. The dilated-layer equipped CNN is applied for replacing the pooling to extract

the large-scale reception fields. Besides applying on UCFCC-50 and the WorldEXPO'10 for people detection, the CSRNet is also used for object detection TRANCOS dataset, and achieved promising results.

The accurate density map of crowd image in one go is not trivial since a number of factors affect the quality of density map like resolution, orientation, image size, quality etc. To address these issues, [149] designed two-level CNN model to achieve the high quality density map at image level. The first level develops a blueprint of the density map whereas the second level make use of the low quality map by mingling up the low-level features create a high resolution density map. Similarly, in [150], the authors introduced a new augmentation approach (see Figure 5) to instruct fully connected CNN for efficient crowd counting. The task of multitask learning is also used to reduce the chances of over-fitting. This also help in generalization of the features across multiple domains. Furthermore, the features extracted from the image are projected to single vector depiction. Significant results were achieved on three different challenging datasets prove the efficacy of the proposed model.

The approach used in [151], [152] relies on the head detection of the persons in the crowd image since the head is regarded as most visible [art of the body]. The head detection is regarded as key task in head counting task that is proposed in the work by using the CNN for counting the people in dense crowd image. A set of comparable results were achieved after performing a series of experiments on UCFCC-50, ShanghaiTech, and AHU-Crowd. A quite new models taht were based on encoder-decoder network called as Scale Aggregation Network (SANet) and Multi-scale Dilated Convolution of Feature Fusion Network (MsDFNet) are developed in [153] and [154] respectively, with the aim to perform accurate and efficient crowd counting. The encoder module tries to explore and retrieve the multi-scale features using the scale aggregation modules while the decoder module creates the high quality density maps by using set of convoluted transposed convolutions.

The major issues in crowd monitoring task including occlusion detection, pose variation and background clutter are addresses in [155] by identifying a person using a the spatial and temporal features with respect to the background and semantics of the scene. The bottom-up CNN of the proposed model has connections to the other top-down CNN that generates the orientation of different poses of the persons of the crowd. Similarly, the authors in [156], designed CNN-based model named ADCrowdNet that analyzed the noisy and congested crowd images for crowd analysis. The purpose of the model is manifold including the crowd counting with the density map generation; occlusions detection and crowd monitoring.

An unsupervised model for crowd counting is designed in [157] that helps in learning a CNN. This model deals the crowd images in a very different way by obtaining a high quality annotated data for dense crowds. To achieve this, Grid WinnerTake-All (GWTA) neural network is modeled to generate supervised data from learning the useful features from unlabeled images. Its is pertinent to mention that about

TABLE II. Statistics of the popular datasets used for crowd counting and density estimation.

Dataset	Number of Images	Average Count	Source of images	Avg. size of image	Annotated persons
ShanghaiTech A [73]	482	500	Streets of Shanghai	584×864	330,165
ShanghaiTech B	716	387	Streets of Shanghai	584×864	241,6775
UCF-CC 50 [50]	50	1279	FLICKR	984×648	64,321
WorldExpo'10 [131]	103	473	Shanghai 2010 WorldExpo	589×86	1132
UCF-QNRF [75]	1535	50	FLICKR, Web Search and Hajj footage	2500×187	1,251,642
JHU-crowd [132]	4,374	117	FLICKR, Shutterstock	1000×750	1.51 million
UCSD [46]	1992	45	UC San Diego	157×237	49,745
Mall [133]	2000	24	Public places	—	60,0005
NWPU-Crowd [134]	5,109	418	Resort, walking street etc	2191×3209	2,133,375

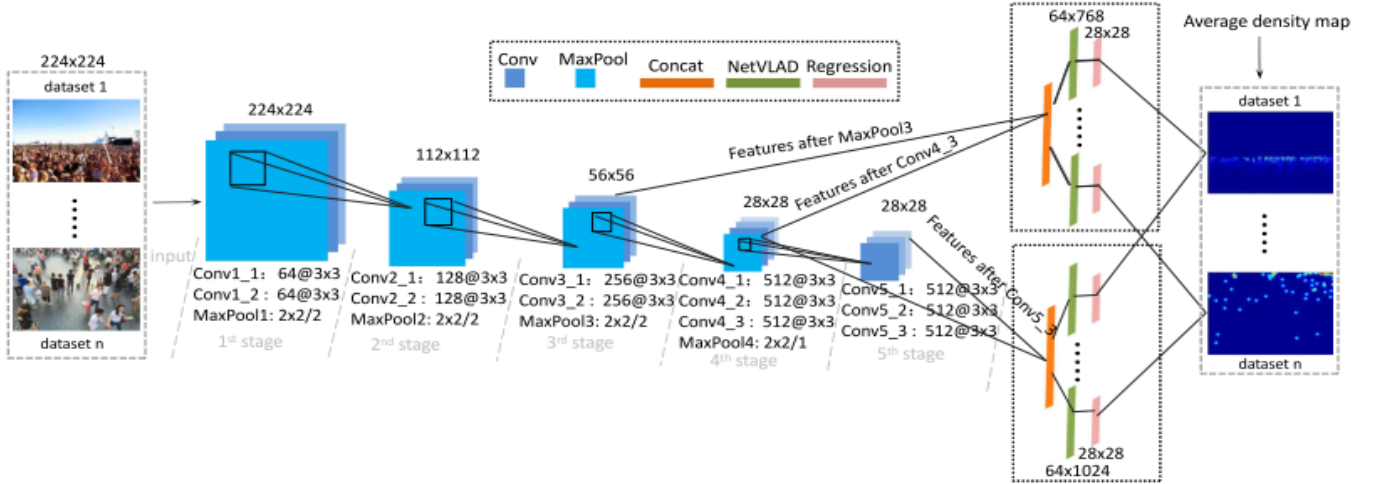


Fig. 5: Architecture of multi-scale multitask deep CNN-based model [150].

99.9% of the tuneable parameters of the GWTA were learned from unlabeled crowd images keeping in view that the remaining parameters were kept updated with principled way of supervision. On a given crowd image, the authors in [158], computed the corresponding density map based on the pixel value depicting where each pixel value of the density map is compatible with respective pixel location in the image. The model is also trained to focus on the the local and global features and scale values. On the basis of these features and scale values, the proposed model outperformed the state-of-the-art methods for crowd counting on a number of standard datasets.

Unlike conventional methods that directly fuse multiple features with weighted average or concatenation, the authors in [159], first introduced a Structured Feature Enhancement Module based on conditional random fields (CRFs) to refine multiscale features mutually with a message passing mechanism. Second, they utilized a Dilated Multiscale Structural Similarity loss to enforce the proposed Deep Structured Scale Integration Network (DSSINet) to learn the local correlation of people's scales within regions of various size, thus yielding high-quality density maps of the crowd image.

A trellis encoder-decoder network (TEDnet) for crowd counting is designed in [160] that assist in creating the density estimation maps for the crowded images. The experiments were performed on four complex benchmark datasets thus

achieving the overall performance in both density map and crowd using MAE as evaluation parameter. Figure 6 shows the comparison of the results with the results obtained through [153] and [73].

Counting the number of people for crowd in the wild is not trivial task due to changeable environment, large-range number of people may cause the current methods can not work well. In addition, due to the scarce data, many methods suffer from serious issues like occlusion etc. To address these issue, in [161], the authors build a large-scale, diverse synthetic dataset. Then they proposes two schemes that exploit the synthetic data to boost the performance of crowd counting in the wild: 1) to pre-train a crowd counter on the synthetic data, then fine-tune it using the real data that significantly prompts the model's performance on real data; 2) to propose a crowd counting method via domain adaptation, which can free humans data from heavy data annotations. Similarly, the work reported in [162], the authors bring forward a novel crowd counting approach that leverages abundantly available unlabeled crowd imagery in a learning-to-rank framework. To induce a ranking of a crowded scene image is guaranteed to contain the same datasets for crowd counting. They collect two crowd scene datasets from Google using keyword searches and query-by-example image retrieval, respectively. They also described how to learn in efficient way by training the network from these unlabeled datasets by incorporating learning-to-

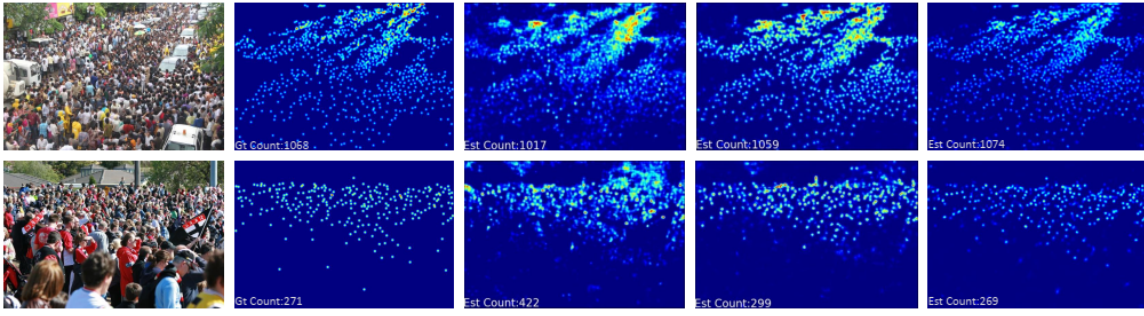


Fig. 6: An illustration of estimated density maps and crowd counts generated by the proposed approach and other state-of-the-arts. The first column shows two samples drawn from ShanghaiTech Part A. The second column shows corresponding ground truth maps with fixed Gaussian kernels. From the third to the last column we show the density maps estimated by MCNN[153] and, SANet [73], and the proposed TEDNet, respectively. TEDnet generates density maps closer to the ground truth, and more accurate crowd counts. [160].

rank in a multi-task network. This learning simultaneously ranks images and estimates crowd density. Experiments on two of the most challenging crowd counting datasets showed that the proposed approach obtains state-of-the-art results.

A Bayesian marked point process (MPP) model is developed in [72] to detect and count people in crowded scenes. The model couples a spatial stochastic process governing number and placement of individuals with a conditional mark process for selecting body shape. The authors automatically make the learning process from the video frames by estimating a mixture of Bernoulli shape prototypes along with an extrinsic shape distribution describing the orientation and scaling of these shapes for any given image location. The reversible jump Markov Chain Monte Carlo framework [163] is used to efficiently search for the maximum a posteriori configuration of shapes, leading to an estimate of the count, location and pose of each person in the scene. The quantitative results achieved for crowd counting are presented for two publicly available datasets with known ground truth. Similarly, the authors [164] proposed a novel CNN-based perspective-guided convolution (PGC) for crowd counting, that aims to overcome the dramatic intra-scene scale variations of people due to the perspective effect. Furthermore, they also created Crowd Surveillance, a large scale dataset for crowd counting that contains 13,000+ high-resolution images with challenging scenarios. The proposed PGCNet is single-column with moderate increase in computation, and extensive experimental results on four benchmark datasets show the improvements of our method against the state-of-the-arts.

A pint-supervised deep detection network is created in [165] for the crowd counting using person detection. The bounding boxes were firstly initialized with the person-head annotation point. These bounding boxes were updated at each iteration during the learning process. The regression method is performed in a way to compare next bounded box with the previous one to stalk the moving pattern of the person. An extensive set of empirical operations were performed for both the detection and counting the persons on a number

of standard datasets like ShanghaiTech, UCF-CC 50, WiderFace, and TRANCOS datasets thus achieving the negotiable results. A quite novel architecture called De-background Detail Convolutional Network (DDCN) is designed by [166], [167], to project the input image to its density map. The proposed model is aimed to remove the involvement of the background from image crowd thus reducing the mapping range. This removing of the background assists in optimizing the learning process. The proposed DDCN works in three phases: in first phase, the decomposer acts as feature extractor by removing the background impression from the crowd image. In second phase, several loss weight functions are induced to compute the Euclidean loss to perform the task of counting. Several experiments were performed on three crowd counting datasets to validate the performance of DCNN.

A perspective-aware CNN (PACNN) for efficient crowd counting is reported in [42] that merges the perspective information into density regression to inform about the person scale-change in the crowd image. For training purpose, the ground truth impressions and maps are created. The purpose of creating PACNN is to detect the change in scale in the multi-scale perspective map. The weights are changed with the scale-change iteratively at every pixel of the density map. The standard datasets like the ShanghaiTech, WorldExpo'10, UCF-CC 50, and UCSD were evaluated using proposed PACNN and demonstrated the efficacy of the model and demonstrate the effectiveness and efficiency of PACNN. A mobile system-based framework is designed by [168] that makes use of smart-phones to capture images to detect the persons in crowd of some designated area. The components of the mobile like clock, camera and GPS are used for proving the additional data like image size, date and time, location (if available), image type and orientation etc. The results are then compared with the standard model namely, You Only Look Once v2 (YOLO2), Aggregate Channel Features (ACF) and Multi-Task cascade CNN (MTCNN) and offered significant results both in detection and counting.

when the scene is too large; i.e. captured in public parks, concerts, stadium or subway platforms; to fit into the view allowed by the field-of-view of the camera, the chances of

occlusion among the large objects. To address these issues, the authors in [169], proposed a deep neural network framework for multi-view crowd counting. The proposed model is made capable to merge the information coming from multi-camera views to sketch a density map at scene level. The proposed model is evaluated on the two benchmark datasets, PETS2009 and DukeMTMC and generate significant results. In a similar work [170], the Distribution Matching for crowd COUNTing (DM-COUNT) is proposed that make use of of Optimal Transport (OT) to compute the difference (or similarity) among the normalized versions of the predicted density map and ground truth density map. A new loss function namely the Total Variation loss is introduced in to preserve the functioning of the OT computation. The experiments show the proposed loss function is asymptotically very tight as compared with that of the Gaussian-based loss function by evaluating the various datasets like UCF-QNRF and NWPU in terms of Mean Absolute Error.

The task of crowd counting based on analyzing the single image-based crowd image attracted a number of researchers specially in crowd image having congested scenes. Such images are analyzed in the work of [171], [172] that provided the concept of Hierarchical Attention-based Crowd Counting Network (HA-CCN). The network is based on vGG16 model and contains spatial attention module (SAM) to modify the the low-level features in the network whereas the second component, global attention module (GAM) works on improving the channel-wise information at higher levels of the network. The proposed method is designed in a way that is easy to implement and analyzes the single crowd image. This new set up reduces the burden of acquiring labor intensive point-wise annotations for new datasets while improving the cross-dataset performance. The proposed model is evaluated on several standard datasets and acquired quite significant results. Figure 7 shws the proposed model. Similar work is also observed in [173] that applied the dense detection framework for crowd counting. The proposed Locate, Size, Count-CNN (LSC-CNN) model is composed of multi-column network with the ability to work in top-down way to detect the persons in the dense crowd. The proposed model apply the bounding box around the located person using point-head annotation. The experiments show that the proposed model not only detects the person in dense crowd but equally applicable also in localization of the occluded objects.

In some noteworthy work [174], [175], a new deep neural network-based model is designed for online crowd monitoring embedded in drone assisted systems. The variant of CNN known as Residual Network (ResNet) is developed that make use of RMSprop algorithm [176] to train the drone system. The results obtained from the experimental work showed accuracy above 90% with significant values of precision and recall. A quite different approach is applied in [8] in which the author presented a locality-aware data partition (LADP) method to combine the training data in different groups on the basis of locality-sensitive hash value. Then LADP assist in developing the balanced data for further processing. Furthermore, a new data augmentation technique is also proposed called as locality-aware data augmentation (LADA) in order

to minimize the learning bias. A hierarchical dense dilated deep pyramid feature extraction through convolution neural network (CNN) for single image crowd counting (HDPF) is designed in [177] to resolve the inherent issues of standard dilated convolution (SDC). The HDPF model is comprised of three modules: general feature extraction module (GFEM) that is responsible of retrieving the general but independent features.; deep pyramid feature extraction module (PFEM) that plays key role in achieving the the relevant contextual information computed by the dense pixel sampling rate. The third component, fusion module (FM) merges the features obtained from other two modules. The proposed technique is evaluated through rigorous simulations on three well known datasets: Shanghaitech (Part-A), Shanghaitech (Part-B) and Venice and the results showed the efficacy of the proposed model in terms of accurate results.

In the density map generated through CNN, each person is depicted through a Gaussian blob. The final count of the image is obtained by computing the integral of map. However, the dense regions may generate inaccurate dense map due to the accumulation of the various Gaussian blobs on small set of pixels. This major issue results in inaccurate analysis of long-tailed pixel-wise distribution map. In order to resolve this issue, the authors in [178], [179] proposed a simple and effective Learning to Scale (L2S) module that is designed in a way to compute the reasonable closeness value of a dense region in the image. Moreover, the proposed L2S also resolve issue of occlusion of blobs by normalizing in different patches. This activity helps significantly in learning the density map. the authors also evaluated the model in localizing people in the image using the local minima of the quantized distance. After performing a series of experiments, the proposed model *AutoScale* provided quality results on some standard datasets. Some key issues related to variation of size in crowd images like occlusion, overlapping etc. are addresses in the work of [180] by developing Multifaceted Attention Network (MAN) to enhance the transformer models in local spatial relation encoding. A number of benchmark crowd counting datasets namely ShanghaiTech, UCFQNRF, JHU++, and NWPU have validated the proposed method.

In order to address the issue of requirement of camera calibration for both the training and testing purpose, the authors [181] proposed the calibration-free multi-view crowd counting (CFMVCC) method. This method aims to computes the scene-level count from the corresponding density map for each camera image without support of the camera calibrations. The proposed CF-MVCC works in a way that at first, it calculates the homography matrix in order to justify each pair of camera-images by estimating the probability or each camera-image pair. In the last phase, by the help of weight maps, the overall count at scene-level is achieved as a simple weighted sum of the density maps for the camera images. In a recent work, the authors [182], advocated estimating people flows across image locations and inferring their densities from these flows to boost performance without requiring a more complex architecture. The authors also showed that leveraging people conservation constraints in both a spatial and temporal manner reduces annotation cost while still leading to similar

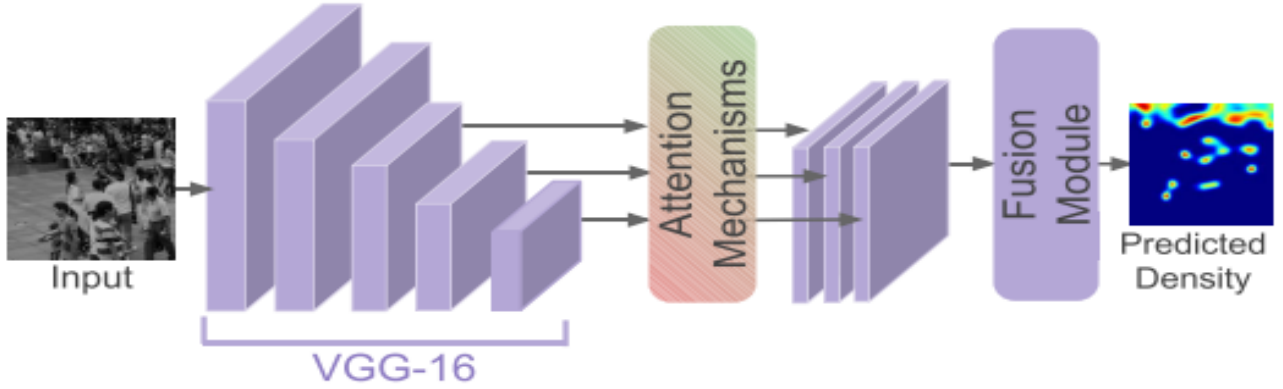


Fig. 7: Overview of the proposed Hierarchical Attention-based Crowd Counting Network (HA-CCN) [171].

performance.

Similarly, in [183], the authors developed an efficient crowd counting neural architecture search (ECCNAS) framework to search efficient crowd counting network structures. A novel search from pre-trained strategy enables the cross-task NAS to explore a large and flexible search space with less search time and get more proper network structures. A novel latency cost estimation algorithm is developed to reduce the computational cost, in terms of multiply-and-accumulate (MACs), by up to 96% with comparable accuracy. Experiments show our searched models get an excellent trade-off between computational complexity and accuracy and have the potential to deploy in practical scenarios with various resource budgets. A lightweight scale-aware network (LSANet) for high-accuracy real-time crowd counting is designed in [184], consisting of four structures: local extractor (LE), scale feature extraction module (SFEM), scale feature fusion module (SFFM), and density map regressor (DMR). The SFEM and SFFM reinforce the learning capability of scale representation, and the MIOU loss function reduces background false recognition and estimation errors. Extensive experiments on four mainstream datasets demonstrate that their proposed method achieves the optimal trade-off between effectiveness and efficiency.

Table III depicts the summary of the papers reviewed published year wise, in this article.

B. Crowd counting at video level

To the best of our knowledge, the CNN was first introduced in the work of [4] in which the authors presented a novel approach to crowd counting that overcomes the challenge of counting people in new surveillance scenes without laborious data annotation. The proposed deep CNN is trained with two related learning objectives, crowd density and crowd count, to achieve superior performance. The proposed deep CNN approach differs from existing crowd counting methods in that it is specifically designed to handle the challenge of cross-scene crowd counting, where no laborious data annotation is required for counting people in new target surveillance crowd scenes unseen in the training set. The deep model has better capability for describing crowd scenes than other hand-craft

features, and our switchable training scheme with two related learning objectives, estimating density map and global count, achieves lower loss by assisting each other. Our approach also includes a data-driven method to fine-tune the pre-trained CNN model adapting to the unseen target scene. The model is evaluated on the three datasets including their proposed dataset namely WorldExpo'10 and two other datasets, UCSD pedestrian dataset [46] and the UCF-CC 50 dataset [50]. Figure 8 depicts the proposed proposed model for cross-scene crowd counting.

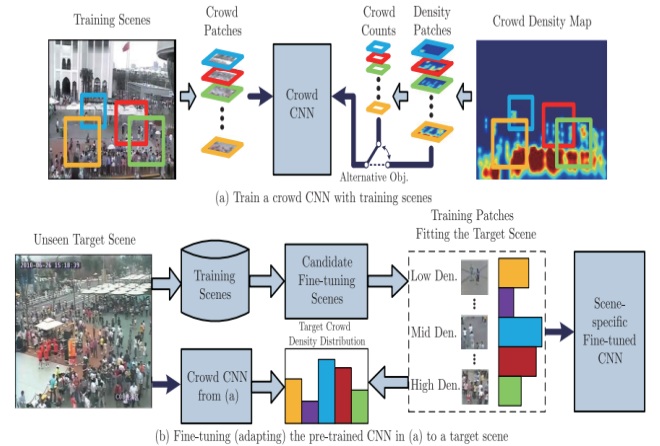


Fig. 8: The proposed cross-scene crowd counting method [4].

Similarly, in [189], the authors claim that their work is pioneer in introducing CNN for the task of crowd density estimation. Furthermore, the authors presented a novel idea by cascading two CNN classifiers with the aim to improve the accuracy and computation speed. For evaluation purpose, the three video clips from the datasets namely, PETS-2009, a Subway image sequence and a ground truth image sequence. The experiments were performed in controlled environment thus confirming significant performance.

A variant of CNN knows as Faster Region-based CNN (F-rCNN) is made into use in [192], [190] in which the authors presented a quite different approach to crowd counting using deep learning techniques. By leveraging the distinguish-

TABLE III. Statistics of the popular datasets used for crowd counting and density estimation.

Year	References	Total
2001	[53], [54], [81], [2], [20]	5
2002	[15], [84], [76], [86]	4
2003	[185], [45], [87], [88]	4
2004	[89], [90], [61]	3
2005	[67], [16], [62], [70]	4
2006	[91], [92], [63]	3
2007	[64], [65], [17], [186]	4
2008	[68], [94], [77], [46], [18]	5
2009	[47], [48], [72], [96], [118]	5
2010	[113], [97], [51]	3
2011	[98], [49], [78], [114]	4
2012	[55], [187], [101]	3
2013	[19], [79], [80], [50], [115], [41]	6
2014	[104], [188], [69], [105]	4
2015	[56], [3], [4], [6], [5], [99], [189], [107], [40], [74], [117], [119], [106], [190]	14
2016	[110], [52], [71], [108], [58], [116], [191], [100], [192], [193], [194], [120], [109], [195], [73], [196], [197], [112], [14], [37], [43]	21
2017	[25], [35], [198], [125], [199], [121], [200], [201], [122], [129]	10
2018	[13], [24], [75], [95], [128], [202], [136], [135], [203], [137], [141], [144], [145], [138], [147], [204], [150], [151], [153], [155], [162]	21
2019	[29], [30], [33], [38], [42], [205], [156], [157], [179], [206], [102], [126], [158], [160], [140], [207], [208], [103], [72], [165], [166] [57], [42], [168], [171], [152]	26
2020	[23], [21], [22], [26], [31], [12], [36], [39], [44], [123], [170], [132], [173], [209], [124], [167], [44]	17
2021	[59], [210], [211], [174], [130], [172], [175]	7
2022	[8], [148], [177], [178], [180], [181], [182], [183], [60], [154], [127], [184]	12
2023	[212], [213], [214], [215], [216], [217], [218]	7

ability of the Convolutional Neural Networks and correlation tracking. Two datasets namely, Oblique and Vertical datasets were evaluated and achieved significant accuracy using Mean Absolute Error (MAE) and Means Square Error (MSE) as evaluating parameters. The researchers also claim that their proposed model is different from the other models by using the ability of Faster R-CNN to detect head-shoulder of each person than estimating the crowd in one frame. Unlike most previous methods which estimate the whole crowd size from one frame. Similarly, the authors in [191] aimed to merge the connotation information related to the moving person with the learning locality-aware feature (LAF) sets [186] for getting the accurate estimation for crowd counting. To achieve this, the CNN in the pixel space is convoluted with densely connected attribute feature map. The feature map indicates the probabilistic strength of a specific semantic class.

The model used for end-to-end crowd monitoring by analyzing the off line videos in [194] is the similar to that was adopted by the authors in [120]. The crowd density estimation is performed by generating and training the model with a new loss function that operates on sets of detection. The conventional model of CNN with two modification is applied in [195] for crowd counting : boosting at layer level sampling with selective features. This result in improving the accuracy and minimize the computation time. Furthermore, the promising feature of this proposed method is that it generates effective output even in noisy data.

A different dataset named TRaffic ANd COngestionS (TRANCOS) [74] is evaluated by using a novel model based on CNN specifically designed for head count named Counting CNN (CCNN) [196]. Then a scale-aware counting model, the Hydra CNN is designed to estimate object densities in a number of crowded scenarios specially when no geometric and background information is available. In [198], the key

factors of related to crowd counting are modeled and extracted from 2D scenic videos. Then a multi-task learning model is proposed to assist in turning the scene models to different sub tasks. These subtasks are then fed to deep CNN for training purpose. These semantic features play key role in understanding the patterns of the moving people and objects in a video. The experimental results on the four state-of-the-art datasets provide significant results when compared with some standard approaches.

A quite useful model based on residual CNN is developed in [200], [210] for performing a number of tasks related to crowd monitoring like crowd counting, detection of the aggressive behavior and classification of the density level according to the crowd strength. The empirical results showed that about 90% accuracy is achieved while detecting the violent behavior. For crowd counting, a number of standard datasets are evaluated and highlighted the superior generalization of crowd analysis. Similarly, in [201], an unsupervised approach for crowd scene anomaly detection and localization is proposed based on a social network model. The video-scene is analyzed using spatial and temporal features by making into use the window-based approach. This activity generates the spatio-temporal cuboids that helps in modeling the crowd behavior using local social networks (LSN). Based on the behavior of the local social networks, a global social network (GSN) is built to depict the global behavior of the overall scene.

The Dilated Convolutions deep network works in sense that it "inflate" the filter kernel by embedding non-zero arbitrary values among the values of the kernel. The widening feature of the kernel is controlled by the dilation parameter of the kernel. This model is used for first time in crowd counting in [202], in which the authors use the multiple columns to extract multiscale information from images framed through

offline video analysis. The proposed model is efficient enough to merge the large scale information gathered from the multi-column structure to improve performance. The results obtained through extensive experiments on the standard datasets like UCSD, UCF and, TRANCOS proved the significance of the proposed model.

The framework proposed in [203] make use of the Recurrent Spatial-Aware Refinement (RSAR) module consisting of two components. One component is responsible to locate the regional the dynamically locates a complimentary region from map formed by the crowd density and then aim to metamorphose it some appropriate scale values for optimal estimation of crowd. While the second component is basically Local Refinement Network (LRN) that performs the refinement of the generated map using residual learning. The proposed model is shown in Figure 9.

The literature [1], [219] conclude that one of the key issues related to crowd counting and density estimation is to analyze the way people appear in crowds. This issue is addressed in [204] using CNN with some modification at hidden layer level that assist in enhancing the ability to count the people by analyzing the crowded images captured from different orientations.

In [206], the author created a first real-world RGB-D People Counting DataSet (PCDS) that contains more than 4500 videos that has the surveillance videos of the passengers entering the buses and trains in some controlled environment. Furthermore, these video were also analyzed for counting the people in an efficient way by computing the point clouds from each frame and transforms to the ground plane. These projected impressions are then carefully examined to detect the potential human heads. The results achieved efficient results and concluded that the proposed model is equally applicable for online crowd counting in intelligent transport systems.

A novel architecture termed as 'temporal channel-aware (TCA)' block is designed in [207], [209] for crowd counting that is trained to achieve the ability to retrieve the temporal and spatial dependencies among video sequences that were analyzed. By stacking TCA blocks together, the authors succeeded to obtain the deep trainable architecture based on CNN, called as enhanced 3D convolutional networks (E3D). For evaluation, extensive experiments were performed on the vehicle dataset TRANCOS thus proving the generalization of the proposed model. Compared with single image based crowd counting, video would help improve the robustness of crowd counting. The orientations like translation, rotation and scaling of people lead to the change of density map of heads between neighboring frames. These issues lead to the change of head counts. To resolve these issues in video crowd counting, a Locality-constrained Spatial Transformer Network (LSTN) is proposed [220] for computing the density map for each frame. Then to relate the density maps between neighboring frames, a Locality-constrained Spatial Transformer (LST) module is introduced to estimate the density map of next frame with that of current frame. To facilitate the performance evaluation, a large-scale video crowd counting dataset is collected, which contains 15K frames with about 394K annotated heads captured from 13 different scenes. Upto the authors best knowledge, it is

the largest video crowd counting dataset among the publicly available datasets. other crowd counting datasets validate the effectiveness of the proposed LSTN for crowd counting.

In [205], the proposed system consists of four modules: people detection, head-torso template extraction, tracking and crowd cluster analysis. Firstly, the system extracts human silhouettes using inverse transform as well as median filter reducing the cost of computing and handling various complex monitoring situations. Secondly, people are detected by their head torso due to less varied and hardly occluded. Thirdly, each person is tracked through consecutive frames using the Kalman filter [221] techniques with Jaccard similarity [222] and normalized cross-correlation [223]. Finally, the template marking is used for crowd counting having cues localization and clustered via Gaussian mapping for normal/abnormal events detection. The experimental results on two challenging datasets of video surveillance such as PETS2009 and UMN crowd analysis datasets demonstrate that the proposed system provides 88.7% and 95.5% in terms of counting accuracy and detection rate.

In a recent work, the authors [215] came with the idea of developing a Feature Pyramid Attention Network (FPANet) with the aim to compute the crowd counting with higher accuracy. The proposed FPANet contains three components: the feature pyramid module (FP), attention module (AM), and multiscale aggregation module (MA). The FP component is made to retrieve the multiscale features from the crowd image to improve the to ability of expressing multiscale. The AM component consists of two inter-related units to focus on the head regions of the persons. It also draws relation among the channel maps to restrains the background information. While the third component AM, integrates the varied scale features to bridge the and cross-channel attention information. Extensive experiments on the benchmark datasets generated the significant results thus proving the importance of the proposed model. Figure 10 depicts the framework of the proposed model.

Similarly, in [214] a pyramidal vision transformer network is developed to counter labeled data in a weakly-supervised way for the end-to-end crowd counting. The concept is applied to a several datasets and compared with results obtained through fully supervised data thus achieving negotiable results.

A number of factors like the noise, distortion, flipped image, quality of the crowd image etc. may affect the quality of density map of the corresponding image, that eventually hinders the task of crowd counting. Although the secondary tasks like binary segmentation may be utilized to assist in enhancing the feature representation, yet it may be isolated as a separate task. In order to counter these issues, the authors in [214] coined the idea of Multi-task Credible Pseudo-label learning (MTCP) model capable of calculating the crowd count and density estimation in three phases. To lessen epistemic vulnerability, perform various tasks learning takes full advantage of directed information in a modest quantity of marked information by managing the named information as per the anticipated confidence map for information expansion. The credible pseudo-labels (CPLs) of density maps are created from an enormous amount of unlabeled information to lighten

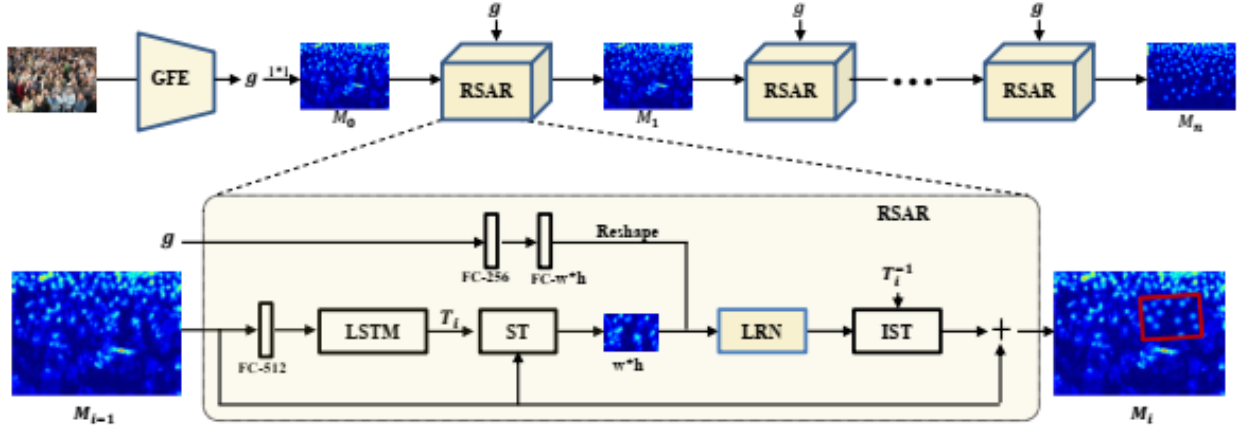


Fig. 9: The architecture of the proposed Deep Recurrent Spatial-Aware Network for crowd counting [203].

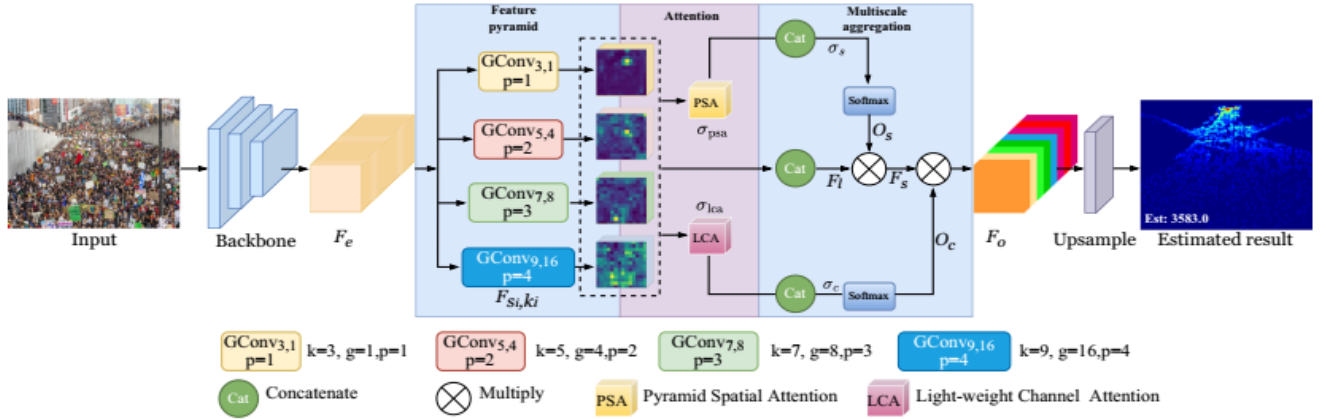


Fig. 10: Diagram of the proposed FPANet for crowd counting [215].

aleatoric vulnerability.

Table IV shows the list of notable journals and conferences that published the papers reviewed in this article.

TABLE IV. Number of articles published (reviewed here) in different journals and conferences.

Journal Name	No. of articles
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	11
International journal of computer vision	6
IEEE Access	3
Applied Intelligence	2
Multimedia Tools and Applications	2
Conference Name	No. of articles
IEE Conference on Computer Vision and Pattern Recognition (CVPR)	46
IEEE International Conference on Image Processing (ICIP)	6
IEEE International Conference on Computer Vision (ICCV)	6
International Conference on Pattern Recognition (ICPR)	6
ACM International conference on Multimedia	3
IEEE Workshop on Performance Evaluation of Tracking and Surveillance	2
IEEE International conference on multimedia and expo (ICME)	2

IV. DATASETS AND EVALUATION PARAMETERS

In this section, a detail discussion on the standard datasets is given on which the researchers performed the experiments for crowd counting and density estimation. Furthermore, the parameters and metrics used for evaluating the applied models are also presented.

A. ShanghaiTech Dataset:

The ShanghaiTech dataset [73] is one of the datasets that were extensively used by the researchers for crowd counting and density estimation. The dataset is mainly divided in two sections namely: Part-A and Part-B. The Part-A is having 482 annotated images and Part-B has a total of 716 images. Both the sections further has *train* and *test* subsets having 300 and 182 images; and 400 and 316 images, respectively. Each crowd image is annotated with close to the center of the head of each person, resulting about 330, 165 annotated persons. Almost all the images are collected from the surveillance cameras located at the busy streets of Shanghai.

B. UCF-CC 50:

In this highly acclaimed dataset, the images contain the extremely dense crowds. A total of 1297 images are collected

from a well-known social platform of images called as FLICKR. The authors officially claim that these images are for only for the research and experimental purposes [50].

C. WorldExpo'10:

This dataset [131] is also divided in two subsections having about 1,127 60-minutes long video files. Out of these, 103 frames were labeled as training and validation. Each of the training video consists of 3 supervised frames. The time interval between the two supervised frames is 15 seconds. It is worthy to note that one has to get official permission from the authors to have access to the dataset.

D. UCF-QNRF:

In this repository, almost all the images are of high resolution that is the reason most of the authors who wants to get high quality density estimation of the crowd images, they make use of this dataset in their experiments. This dataset has about 1535 images having crowded people impressions with annotated 1,251,642 persons. This is one of the datasets which is highly recommended for crowd counting and density estimations from a number of authors.

E. JHU-crowd:

This multi-variant dataset has 4,250 images with more than 1 million annotations [132]. This dataset is regarded as one of the richest and comprehensive collection that has a variety of different scenarios and environmental conditions including a number of weather-based scenarios and illumination variations. This diversity makes the dataset a very challenging and complex option for the researchers.

F. UCSD:

The UCSD Crowd Counting dataset [46] was collected from a stationary surveillance camera with the purpose of security measures to look over the passing by pedestrian on the side walks. In this dataset, the images show both the varieties of the crowd i.e. from sparse to dense. It contains about 1900 images of persons in crowd.

G. Mall:

In this dataset [133], the video frames in jpeg format, the ground truth values including the perspective normalized features, and perspective normalization map etc. are included. The dataset was collected from a publicly accessible web-cam for crowd counting and profiling research. Over 60,000 pedestrians were labeled in 2000 video frames. The authors annotated the data exhaustively by labeling the head position of every pedestrian in all frames.

H. Evaluation parameters

The evaluation parameters that were mostly recommended in crowd counting and density estimation literature are, Mean Absolute Error (MAE) and Mean Square Error (MSE). The formal definitions of these metrics are given below:

$$MAE = \frac{1}{N} \sum_{i=1}^N |x_i - x'_i| \quad (1)$$

$$MAE = \sqrt{\frac{1}{N} \sum_{i=1}^N |x_i - x'_i|^2} \quad (2)$$

here N depicts the number of images being tested, x_i is representing the image number i from the test set and x'_i is the estimated counting of image i . In data science and applied statistics, MAE is measured as the average distance between the real data and the predicted data. Lower the values of MAE, higher will be the rate of prediction. Similarly, MSE computes the average squared difference between the predicted values and the actual values in a sample space.

Some of the authors [224], [225], [35], [73], [226] used region of interest (ROI) metric in crowd counting that refers to a specific area within an image or a scene where the crowd count estimation is desired. The ROI is typically selected based on the specific application or the requirements of the task. The choice of ROI depends on various factors, including the camera placement, the scene layout, and the specific objectives of the crowd counting system. The ROI is usually defined to encompass the area where the crowd is present or where the accurate estimation of crowd density is needed.

When determining the ROI, it is important to consider the following factors:

Camera Placement: The camera should be positioned in such a way that the desired crowd area is within its field of view. The ROI can be selected to cover the entire field of view or a specific portion of it.

Scene Characteristics: Understanding the scene and its characteristics is crucial for selecting an appropriate ROI. Factors like crowd distribution, crowd density variations, and areas of interest within the scene influence the choice of ROI.

Task Requirements: The specific objectives of the crowd counting task can guide the selection of the ROI. For example, if the goal is to count the number of people entering a specific entrance or passing through a particular area, the ROI can be defined to cover that specific region.

It's worth noting that in some cases, the entire scene may serve as the ROI, especially when the goal is to estimate the overall crowd density or monitor crowd movement across the entire area captured by the camera. Ultimately, the selection of the ROI in crowd counting is a design decision that depends on the particular application and the requirements of the task at hand.

Table V shows the comparison of the results on benchmark datasets obtained through CNN-based methods using MAE and MSE as evaluation criteria.

V. CONCLUSION

In applied machine vision, the task of crowd counting is one of the challenging and complex problems that persuaded a number of researchers to apply conventional and deep learning-based algorithms to acquire significant results. In this paper, we cover the noteworthy articles related to crowd counting and density estimation published last 23 years. Furthermore, we have divided the collected literature in two broad sections i.e. one section will describe the work in which conventional approaches are used while the other section will elaborate the work based on CNN-based algorithms. The literature in which conventional based approaches are used are further split in two divisions based on the type of the target used for analysis i.e. image or video. Furthermore, the details about the benchmark datasets that were used for experiments for getting empirical results is given. Table V shows the summary of the results obtained on different datasets the basis of the evaluation parameters on which the results are tested and analyzed. In addition, the papers that are reviewed in this article are also shown in Table III according to the publishing year. The survey paper covers the design considerations and recent advances with respect to single image crowd counting problem, and summarizes more than 200 crowd counting schemes using deep learning approaches proposed since 2000. We have discussed the major datasets, performance metrics, design considerations, techniques, and representative schemes to tackle the problem. We provide a comprehensive overview and comparison of three major design modules for deep learning in crowd counting, deep neural network design, loss function, and supervisory signal. The research field of crowd counting is rich and still evolving.

VI. FUTURE DIRECTIONS

From the literature review of the field of crowd counting, we conclude that the authors should focus in the following issues in their future work.

- **Improved accuracy:** The researchers should work in developing the algorithms that are able to generate more accurate results and keep pace with the development and enhancements in computer vision and deep learning techniques. It involves investigating the deep network architectures with novel features by encompassing the contextual information about the scenarios thus achieving the generalization of the network.
- **Real-time crowd counting and density estimation:** There is a growing demand for real-time crowd counting systems that can provide instantaneous crowd density estimates. Future work will likely concentrate on developing efficient algorithms and optimizing computational resources to enable real-time processing on various platforms, including embedded systems and edge devices.
- **Multi-view and multi-modal approaches:** Crowd counting can benefit from utilizing multiple views and modalities, such as combining information from multiple cameras or incorporating data from different sensors (e.g., thermal imaging). Future research may explore how to

effectively fuse these diverse data sources to enhance crowd counting accuracy and robustness.

- **Benchmark datasets and evaluation metrics:** Building standardized benchmark datasets and evaluation metrics is crucial for comparing different crowd counting algorithms. Future work will involve creating more diverse and challenging datasets, covering various scenarios, crowd densities, and environmental conditions. Additionally, developing comprehensive evaluation metrics that consider both accuracy and efficiency will be essential for fair performance assessment.
- **Privacy and ethical considerations:** As crowd counting often involves capturing and analyzing visual data from public spaces, future work will focus on addressing privacy concerns and ethical considerations. This may involve developing privacy-preserving crowd counting techniques, adhering to data protection regulations, and ensuring transparency in how the technology is used.
- **Generalization and adaptation:** Crowd counting algorithms should be capable of generalizing across different scenes and adapting to changing environments. Future research may explore techniques for domain adaptation, transfer learning, and robustness against variations in lighting conditions, camera viewpoints, and crowd dynamics.
- **Real-world deployment and applications:** Future work will involve applying crowd counting techniques to various real-world applications, such as crowd management in public spaces, transportation systems, event planning, and safety monitoring. Research will focus on addressing the practical challenges of deploying these systems, including scalability, reliability, and integration with existing infrastructure. The pre-trained models and the crowd attribute predictors have a number of interesting applications in the future work, such as crowd video retrieval and crowd event detection.
- **Data annotation:** There should be a mechanism in data annotation of crowd videos that is based on a query labeler to annotate the the less confident regions. This query based intelligent labeler will significantly minimize the labeling time for the task of crowd counting in videos. Furthermore, this activity will surely help in increasing the model generalization.
- **Estimation from crowd flow:** There is need to develop a system that analyze the temporal patterns from the flow of the crowd in order to estimate the number of people. Instead of applying regressive approaches, the consecutive images may be analyzed for estimating flow across the specific locations in the images. As a result, the annotation cost will be highly reduced to get the more impressive results.

By focusing on these areas, researchers and practitioners can advance the field of crowd counting, enabling more accurate and efficient crowd analysis and facilitating the development of applications that benefit society.

TABLE V. Comparison of conventional and CNN-based methods on different datasets using Mean Absolute Error (MAE) and Mean Square Error (MSE) as evaluation parameters

Datasets	Model (MAE ↓; MSE ↓)
ShanghaiTech A	MCNN [73] (110.2; 173.2); Cascade CNN [125] (101.3; 152.4); Contextual Pyramid CNN (CP-CNN) [199] (73.6; 106.4); Multi-Scale CNN (MSCNN) [121] (83.8; 127.4); Switch-CNN [122] (90.4; 135.0); Deeply-Recursive network (DRResNet) [24] (86.3; 124.2); Adversarial Cross-Scale Consistency Pursuit (ACSCP) [137] (75.7; 102.7); scale-adaptive CNN (SaCNN) [141], [208] (86.8 ; 139.2); De-correlated ConvNet (D-ConvNet) [144] (101.7; 152.8); Fully Connected CNN (F-CNN) [145] (67.6; 110.6); Congested Scene Net (CSRNet) [147] (75.81; 120.82); Incrementally Growing CNN (IG-CNN) (72.5; 118.2); Iterative Counting CNN (ic-CNN) [149] (68.5; 116.2); Multiscale Multitask CNN (MMCNN) [150] (107.6; 169.3); MS-GAN [139] (96.5; 135.9); CNN [151] (1265; 173.5); Scale Aggregation Network (SANet) [153] (67.0; 104.5); Top-Down Feedback (TDF-CNN) [155] (20.7; 32.8); Attention Map Generator (AMG) [156] (70.9; 115.2); Grid Winner-Take-All (GWTA-CCNN) [157] (154.7; 229.4); Attention Map Generator (AMG) [156] (7.7; 12.9); Context Aware Network (CAN) [126] (62.3; 100.0); Deep Structured Scale Integration Network (DSSINet) [159] (6.63; 96.04); Trellis Encoder-Decoder Net (TEDNet) [160] (64.2; 109.1); SPANet [103] (59.4; 92.5); Congested Scene Recognition Net (CSRNet) [148] (63.3; 100.9); Deep Stacked Dilated Convolution(DSDC) [177] (69.9;106.9); AutoScale [9] (65.8, 112.1); Efficient Crowd Counting Neural Architecture Search (ECCNAS) [183] (62.0; 110.9); Hierarchical adaptive regression (HARnet) [60] (52.3; 87.5); Locality-Aware Crowd Counting (LACC) [175] (65.8; 103.5); Multi-scale Dilated Convolution of Feature Fusion Network (MsDFNet) [154] (66.3; 107.1); Lightweight Scale-Aware Network (LASNet) [184] (66.1; 110.2); Attentive multi-stage CNN (AMCNN) [123] (76.1, 110.7); DM-Count [170] (59.7; 95.7); Hierarchical Attention-Based CNN (HA-CNN)[171] (62.9; 94.9); JHU-CROWD [7] (60.2; 94.00); Locate, Size and Count CNN (LSC-CNN) [11] (64.4; 117.0); Automatic Multi-Scale Network (AMSN) [124] (56.7; 93.4); ResNet-DC [210] (73.51; 118.1); Dense Scale Network (DSN) [130] (61.3; 97.3); AutoScale [9] (65.6, 109.3); Domain-guided Channel Attention Network (DCANet) [172] (57.5; 94.9); Perspective-guided Convolution (PGC) [164], [42] (58.3; 90.2); Point-Supervised Deep Detection Network (PSDDN) [165] (65.9; 112.3); De-background Detail CNN (DDCNN) [166] (71.5; 110.4); CSRNet [57] (63.1; 96.2); CAFNet [212] (59.4; 94.5); Directional Context Attention (DCANet) [218] (66.0; 105.2); FPNNet [215] (70.9; 120.6);
ShanghaiTech B	MCNN [73] (26.4; 41.3); DCNN [198] (20.2; 35.6); Cascade CNN [125] (20.0; 31.1); Contextual Pyramid CNN (CP-CNN) [199] (20.1; 30.1); Multi-Scale CNN (MSCNN) [121] (17.7; 30.2); Switch-CNN [122] (21.6; 33.4); Deeply-Recursive network (DRResNet) [24] (14.52; 21.0); Adversarial Cross-Scale Consistency Pursuit (ACSCP) [137] (17.2; 27.4); scale-adaptive CNN (SaCNN) [141], [208] (16.2 ; 25.8); De-correlated ConvNet (D-ConvNet) [144] (25.7; 38.6); Fully Connected CNN (F-CNN) [145] (10.17; 18.8); Congested Scene Net (CSRNet) [147] (10.6; 16.0); Incrementally Growing CNN (IG-CNN) (13.6; 21.1); Iterative Counting CNN (ic-CNN) [149] (10.7; 16.0); Multiscale Multitask CNN (MMCNN) [150] (21.4; 33.9); MS-GAN [139] (18.7; 30.5); CNN [151] (23.76; 33.12); Scale Aggregation Network (SANet) [153] (8.4; 13.6); Top-Down Feedback (TDF-CNN) [155] (97.5; 145.1); Context Aware Network (CAN) [126] (7.80; 12.2); Scale Aware Net(SaNET) [158] (16.86; 28.41); Deep Structured Scale Integration Network (DSSINet) [159] (6.85; 10.34);

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TABLE V – continued from previous page

Datasets	Model (MAE ↓; MSE ↓)
	Trellis Encoder-Decoder Net (TEDNet) [160] (8.2; 12.8); SPatial Awareness Network (SPANet) [103] (6.50; 9.90); Congested Scene Recognition Net (CSRNet) [148] (7.9; 12.2); Deep Stacked Dilated Convolution(DSDC) [177] (9.40;15.9); AutoScale [9] (8.6, 13.9); Multifaceted Attention Network (MAN) [180] (56.8; 90.3); Efficient Crowd Counting Neural Architecture Search (ECCNAS) [183] (7.5; 12.9); Hierarchical adaptive regression (HARnet) [60] (5.96; 9.4); Locality-Aware Crowd Counting (LACC) [175] (8.6; 14.6); Multi-scale Dilated Convolution of Feature Fusion Network (MsDFNet) [154] (8.3; 13.9) Lightweight Scale-Aware Network (LASNet) [184] (8.6; 13.9) Attentive multi-stage CNN (AMCNN) [123] (15.3, 27.4); DM-Count [170] (7.4; 11.8); Hierarchical Attention-Based CNN (HA-CNN)[171] (8.1; 13.4); JHU-CROWD [7] (7.5; 12.1); Locate, Size and Count CNN (LSC-CNN) [11] (8.14; 12.7) Automatic Multi-Scale Network (AMSNet) [124] (6.7; 10.2); ResNet-DC [210] (13.3; 22.5); Dense Scale Network (DSN) [130] (6.74; 10.48); Domain-guided Channel Attention Network (DCANet) [172] (6.5; 10.6); Perspective-guided Convolution (PGC) [164], [42] (9.0; 14.2); Point-Supervised Deep Detection Network (PSDDN) [165] (9.1; 14.2); De-background Detail CNN (DDCNN) [166] (13.8; 20.1); CSRNet [57] (8.72; 13.26); CAFNet [212] (6.8; 11.2); Directional Context Attention (DCANet) [218] (8.5; 14.5); Feature Pyramid Attention Network (FPANet) [215] (8.8; 15.5);
UCF-CC 50	CNN [4] (467.0; 498.5); CNN [5] (8.5, 38); Fully Connected CNN [112] (452.5); –; MCNN [73] (377.6; 509.1); CNN Boosting [195] (376.2; –); CCNN [196] (488.67; 646.68); DCNN [198] (409.55; 563.7); Cascade CNN [125] (322.8; 397.9); Contextual Pyramid CNN (CP-CNN) [199] (295.8; 320.9); Multi-Scale CNN (MSCNN) [121] (363.7; 468.4); Resnet [200] (1150; 1497); ConvLSTM [35] (1.13; 1.43); Switch-CNN [122] (318.1; 439.2); Deeply-Recursive network (DRResNet) [24] (291.0; 404.6); scale-adaptive CNN (SaCNN) [141], [208] (314.9 ; 424.8); De-correlated ConvNet (D-ConvNet) [144] (354.1; 443.7); Fully Connected CNN (F-CNN) [145] (253.1; 356.4); Congested Scene Net (CSRNet) [147] (266.1; 397.5); Incrementally Growing CNN (IG-CNN) (291.4; 394.4); Iterative Counting CNN (ic-CNN) [149] (260.9; 365.5); Multiscale Multitask CNN (MMCNN) [150] (396.3; 564.2); MS-GAN [139] (345.7; 418.3); CNN [151] (338.6; 424.5); Scale Aggregation Network (SANet) [153] (258.4; 334.9); Top-Down Feedback (TDF-CNN) [155] (354.7; 491.4); Attention Map Generator (AMG) [156] (273.6; 362.0); Grid Winner-Take-All (GWTA-CCNN) [157] (433.7; 583.3); Context Aware Network (CAN) [126] (212.2; 243.7); Scale Aware Net(SaNET) [158] (271.6; 391.00); Deep Structured Scale Integration Network (DSSINet) [159] (216.9; 302.4); Trellis Encoder-Decoder Net (TEDNet) [160] (294.4; 354.4); SPatial Awareness Network (SPANet) [103] (232.6; 311.7); SE-GAN (Synthetically Enhanced) (373.4; 528.8); Congested Scene Recognition Net (CSRNet) [148] (267.8; 348.9); Efficient Crowd Counting Neural Architecture Search (ECCNAS) [183] (233.2; 293.8); Hierarchical adaptive regression (HARnet) [60] (181.2; 254.0); Multi-scale Dilated Convolution of Feature Fusion Network (MsDFNet) [154] (183.6; 236.2) Pyramid-dilated deep CNN (PDCNN) [127] (221.1.; 326.6); Attentive multi-stage CNN (AMCNN) [123] (272.5, 387.5); DM-Count [170] (211.0; 291.5); Hierarchical Attention-Based CNN (HA-CNN)[171] (256.2; 384.4); Automatic Multi-Scale Network (AMSNet) [124] (208.4; 297.3); ResNet-DC [210] (254.78; 326.10); Domain-guided Channel Attention Network (DCANet) [172] (57.5; 94.9); Domain-guided Channel Attention Network (DCANet) [172] (182.6; 268.9);

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TABLE V – continued from previous page

Datasets	Model (MAE ↓; MSE ↓)
	Perspective-guided Convolution (PGC) [164], [42] (278.6; 403.5); Point-Supervised Deep Detection Network (PSDDN) [165] (359.4; 514.8); De-background Detail CNN (DDCNN) [166] (286.2; 479.6); FPANet [215] (159.5; 218.4);
WorldExpo'10	End-to-End CNN [120] (11.7, –); MCNN [73] (11.6; –); DCNN [198] (10.5; –) Contextual Pyramid CNN (CP-CNN) [199] (8.86; –); ConvLSTM [35] (10.7; –); Switch-CNN [122] (9.4; –); Deeply-Recursive network (DRResNet) [24] (1.04; 1.35); De-correlated ConvNet (D-ConvNet) [144] (9.1; –); Generative Adversarial Network (GAN) [138] (46.6; 39.5); Incrementally Growing CNN (IG-CNN) (8.9; –); Iterative Counting CNN (ic-CNN) [149] (8.8; –); Multiscale Multitask CNN (MMCNN) [150] (10.3; –); Scale Aggregation Network (SANet) [153] (8.2; –); Top-Down Feedback (TDF-CNN) [155] (11.5; –); Attention Map Generator (AMG) [156] (7.3; –); Context Aware Network (CAN) [126] (7.2; –); Deep Structured Scale Integration Network (DSSINet) [159] (6.67; –); Dual-Goal Generative Adversarial Net (DG-GAN) [140] (8.32; –); SPatial Awareness Network (SPANet) [103] (7.7; –); SE-GAN (Synthetically Enhanced) (26.3; –); Congested Scene Recognition Net (CSRNet) [148] (7.9; –); Locate, Size and Count CNN (LSC-CNN) [11] (225.6; 302.7); Automatic Multi-Scale Network (AMSNet) [124] (6.8; –); Dense Scale Network (DSN) [130] (7.7; –); Domain-guided Channel Attention Network (DCANet) [172] (6.1; –); CSRNet [57] (8.5; –); De-background Detail CNN (DDCNN) [166] (9.8; –); FPANet [215] (6.9; –);
UCF-QRNF	Context Aware Network (CAN) [126] (107.0; 183.0); Deep Structured Scale Integration Network (DSSINet) [159] (99.1; 159.2); Trellis Encoder-Decoder Net (TEDNet) [160] (113.0; 118.0); Multifaceted Attention Network [180] (77.3; 131.5); SE-GAN (Synthetically Enhanced) (230.4; 384.5); Efficient Crowd Counting Neural Architecture Search (ECCNAS) [183] (91.2; 158.9); Hierarchical adaptive regression (HARnet) [60] (83.4; 138.2); Locality-Aware Crowd Counting (LACC) [175] (113.9; 210.5); Pyramid-dilated deep CNN (PDCNN) [127] (117.8; 196.5); Lightweight Scale-Aware Network (LASNet) [184] (86.7; 146.0) DM-Count [170] (85.6; 148.3); Hierarchical Attention-Based CNN (HA-CNN) [171] (118.1; 180.4); JHU-CROWD [7] (95.5; 164.3); Locate, Size and Count CNN (LSC-CNN) [11] (218.3; 8.6); Automatic Multi-Scale Network (AMSNet) [124] (101.8; 163.2); Domain-guided Channel Attention Network (DCANet) [172] (81.0; 147.00); CAFNet [212] (96.4; 155.5); Directional Context Attention (DCANet) [218] (109.8; 187.6);
JHU-Crowd	Multifaceted Attention Network (MAN) [180] (53.4; 209.9);
UCSD	Random Forest [3] (1.61; 4.40); CNN [4] (1.60; 3.31); IIS-LDL [106] (2.08; 7.25); Holistic system [71] (1.84; 5.21); Sparse Linear Regression (SLR) [58] (1.83; 5.04); CNN [116] (0.6040; –); Random Projection Forest (RPF) [191] (1.90; 6.01);
	Visual Object Counting (VOC) [109] (1.98; –); CNN Boosting [195] (1.15; –) MCNN [73] (1.07; 1.35); CCNN [196] (1.65; 1.79); DCNN [198] (1.0; 1.4) Switch-CNN [122] (1.61; 2.10); Congested Scene Net (CSRNet) [147] (1.07; 1.35); Scale Aggregation Network (SANet) [153] (1.02; 1.29); SVR(Boosting DALSVR) [102] (2.10; 1.29); Dual-Goal Generative Adversarial Net (DG-GAN) [140] (0.93; 1.17); SPatial Awareness Network (SPANet) [103] (1.00; 1.08); Multi-Level Locality-Constrained Spatial Transformer Network (MLSTN) [209] (1.02; 1.32); Locality-constrained Spatial Transformer Network (LSTN) [220] (1.07; 1.39); FPANet [215] (76.0; 120.0);
Mall	Random Forest [3] (2.50; 10.0); IIS-LDL [106] (2.69; 12.1); Sparse Linear Regression (SLR) [58] (3.23; 15.77); Random Projection Forest (RPF) [191] (3.22, 15.5); CNN Boosting [195] (2.43; –); SVR(Boosting DALSVR) [102] (9.57; 2.40); Scale Aware Net(SaNET) [158] (1.28; 1.68); Dual-Goal Generative Adversarial Net (DG-GAN) [140] (1.64; 2.13); Attentive multi-stage CNN (AMCNN) [123] (2.35, 3.00); Multi-Level Feature Fusion Based Locality-Constrained Spatial Transformer Network (MLSTN) [209] (1.80; .42); Locality-constrained Spatial Transformer Network (LSTN) [220] (2.00; 2.50); CAFNet [212] (1.91; 2.44);
NWPU-Crowd	AutoScale [9] (85.6, 356.1); MANNet [180] (76.5; 323.0); FPANet [215] (97.1; 372.8);

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